INCH-POUND
MIL-R-62197A(AT)
31 May 1990
SUPERSEDING
MIL-R-62197(AT)
28 October 1983

MILITARY SPECIFICATION

RELAY, ARMATURE

This specification is approved for use by the U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a single pole, double throw (SPDT), hermetically sealed, electromagnetic, direct current, balanced armature relay used in the stabilization system of the M60A1 tank.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extend specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive Command, ATTN: AMSTA-GDS, Warren, MI 48090, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 1250

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-R-5757	- Relay, Electromagnetic, General Specifications for.

STANDARDS

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

11608300	- Controller Unit, Electronic.
12251725	- Relay, Armature.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), relay shall be subjected to first article inspection (see 4.4).

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Material shall be free from defects which adversely affect performance or serviceability of the finished product.

3.2.1 Recycled, virgin, and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.4.2).

3.3 Construction. The relay shall be a SPDT, hermetically sealed, electromagnetic, direct-current, balanced-armature relay, defined as a Category III relay in MIL-R-5757, and shall conform to Drawing 12251725.

3.3.1 Corrosion resistance. Exposed surfaces shall be of corrosion-resistant material or suitably treated to prevent corrosion.

3.3.2 Case. The relay case shall be hermetically sealed per symbol 4 of MIL-R-5757.

3.3.3 Contact arrangement. The relay contact arrangement shall be a SPDT configuration conforming to forming to form C of MIL-R-5757.

3.3.4 Terminals. The relay shall be provided with solder terminals located as shown in Drawing 12251725.

3.3.5 Weight. The weight of the relay shall be not more than 3 ounces.

3.4 Performance.

3.4.1 Insulation resistance. Insulation resistance shall be not less than 1,000 megaohms, with 500 volts, direct-current (vdc) applied (see 4.7.3).

3.4.2 Dielectric strength. The relay shall withstand the specified test voltage without evidence of damage due to arcing, flashover, or insulation breakdown. Leakage current shall not exceed 1.0 milliamperes (mA) (see 4.7.4).

Test Voltage:

(1) Across open contacts – 500 vdc/300 volts, root mean square (vrms).

(2) Other than across open contacts – 1500 vdc/1000 vrms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

3.4.3 Contact characteristics.

3.4.3.1 Rated load. The contacts shall withstand 2.0 amperes (A) resistive at 28 vdc (see 4.7.5.1).

3.4.3.2 Resistance. The contact resistance shall be not greater than 0.001 ohms (see 4.7.5.2).

3.4.3.3 Life. The relay contacts shall be capable of 100,000 operations at rated load. After 100,000 operations, the relay shall meet the contact resistance requirement of 3.4.3.2 (see 4.7.5.3).

3.4.4 Coil characteristics.

3.4.4.1 Current. The relay coil shall have the following characteristics:

Pull in:	Not more than 3.6 mA dc
	Not less than 2.5 mA dc
Drop out:	Not more than 1.8 mA dc
	Not less than .36 mA dc

The difference between pull in and drop out current shall be not less than 0.7 mA dc (see 4.7.6.1).

3.4.4.2 Voltage. The relay coil shall be capable of withstanding continuous application of 18.0 vdc with no adverse effects (see 4.7.6.2).

3.4.5 DC resistance. The coil dc resistance shall be 2000 $\pm$ 200 ohm at standard ambient temperature conditions (see 4.7.7).

3.4.6 Inductance. The coil inductance shall be not more than 25 Henries (H) at 60 Hertz (Hz) (see 4.7.8).

3.4.7 Environmental conditions.

3.4.7.1 Temperature. The relay shall meet the requirements of this specification at ambient air temperatures ranging from -25° Fahrenheit (F) to 145°F. The units shall not be damaged by storage conditions ranging from -65°F to 160°F (see 4.7.9.1.1 and 4.7.9.1.2).

3.4.7.2 Vibration. The relay shall be capable of withstanding the vibration profile per figure 1 for a period of 80 minutes (including up to four resonance dwells of 13-1/3 minutes each) in each of the three mutually perpendicular axes. Opening of closed contacts, while energized, shall not exceed 10 milliseconds (ms) (see 4.7.9.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

3.4.7.3 Shock. The relay shall be capable of withstanding exposure to three half-sine wave shock pulses of 40 ± 4 gravity force (g) for a duration of 18 ± 3 ms applied in both directions along three mutually perpendicular axes (see figure 2). In addition, the relay shall be capable of withstanding three half-sine wave shock pulses of 180 ± 18 g for a duration of 1.7 ± 0.2 ms in both directions along the longitudinal and traverse axes and three half-sine wave shock pulses of 380 ± 38 g for a duration of 1.0 ± 0.1 ms in both directions along the vertical axis. Opening of closed contacts while energized shall not exceed 10 ms (see 4.7.9.3.1 and 4.7.9.3.2).

3.4.7.4 Humidity. The relay shall meet the requirements of this specification when subjected to humidity conditions up to 99 percent relative humidity (see 4.7.9.4).

3.5 Identification and marking. Identification and marking of the assembly shall be in accordance with the requirements of MIL-STD-130.

3.6 Workmanship. Workmanship shall conform to requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, it does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).
c. Control tests (see 4.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed under the following standard (room) ambient conditions:

a. Temperature: 77 ± 18°F.
b. Relative humidity: Uncontrolled room ambient.
c. Atmospheric pressure: Site pressure.

4.4 First article inspection. First article inspection shall be performed on preproduction and initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply relays that are fully representative of those inspected as a first article sample. Any changes or deviations of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two relays. Preproduction inspection shall consist of inspection as specified in table I and the examination of 4.5.2.

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two from the first ten relays produced under the production contract for initial production inspection. The initial production sample shall be inspected as specified in table I, and examined as specified in 4.5.2.

4.4.3 First article inspection failure. Deficiencies found during, or as result of, the first article inspection shall be caused for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, the first article inspection shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

4.5 QCI. QCI shall include the examination of 4.5.2 and the tests of 4.5.3. Noncompliance with any of the specified requirements in sections 3 and 5 shall be cause for rejection of the sample and the inspection lot.

4.5.1 Sampling plan. Unless otherwise specified (see 6.2), the sampling plan specified herein shall be used. See 6.4 for definitions of sampling inspection terms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of inspection.

Title	Requirement	Test	First article		QCI		Control test	
			PPI	IPI	Exam	Test	(1/50)	(1/200)
Materials and construction	3.2 and 3.3	4.7.1	X	X				
Defects (see table III)	3.3 thru 3.3.5, 3.5 and 3.6	4.7.2	X	X	X			
Insulation resistance	3.4.1	4.7.3	X	X				
Dielectric strength	3.4.2	4.7.4	X	X		X		
Contact characteristics	3.4.3	4.7.5						
Rated load	3.4.3.1	4.7.5.1	X	X		X		
Resistance	3.4.3.2	4.7.5.2	X	X		X		
Life	3.4.3.3	4.7.5.3	X	X				X
Coil characteristics	3.4.4	4.7.6						
Current	3.4.4.1	4.7.6.1	X	X		X		
Voltage	3.4.4.2	4.7.6.2	X	X				
DC resistance	3.4.5	4.7.7	X	X		X		
Inductance	3.4.6	4.7.8	X	X		X		
High temperature	3.4.7.1	4.7.9.1.1	X	X			X	
Low temperature	3.4.7.1	4.7.9.1.2	X	X			X	
Vibration	3.4.7.2	4.7.9.2	X	X			X	
Shock	3.4.7.3	4.7.9.3						
Basic design	3.4.7.3	4.7.9.3.1	X	X			X	
High intensity	3.4.7.3	4.7.9.3.2	X	X				
Humidity	3.4.7.4	4.7.9.4	X	X				X

4.5.1.1 Lot formation. An inspection lot shall consist of all relays of a single type, class, style and part identification number (PIN), from an identifiable production period, from one manufacturer, from one manufacturing location, submitted at the same time for acceptance.

4.5.1.2 Sample. The sample for QCI examination shall be randomly selected from the inspection lot in accordance with table II.

4.5.2 Examination. The sample selected in accordance with 4.5.1.2 shall be examined and defects classified as specified in table III (see 4.7.2). The acceptance number in all cases is zero.

4.5.3 Tests. Each relay shall be subjected to the tests specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

TABLE II. Sampling plan for examination.

Inspection lot size	Sample size for examination	
	Major	Minor
2 to 8	*	5
9 to 15	13	5
16 to 25	13	5
26 to 50	13	5
51 to 90	13	7
91 to 150	13	11
151 to 280	20	13
281 to 500	29	16
501 to 1200	34	19
1201 to 3200	42	23
3,201 to 10,000	50	29
10,001 to 35,000	60	35
35,001 to 150,000	74	40
150,001 to 500,000	90	40
500,001 and over	102	40

* Indicates entire lot must be inspected (100% inspection).

TABLE III. Classification of defects.

Category	Defects	Method of inspection
Major		
101	Incorrect dimensions, affecting interchangeability (see 3.3).	SIE 1/
102	Improper weight (see 3.3.5).	SIE
103	Improper marking (3.5).	Visual
104	Improper workmanship, affecting performance (see 3.6).	Visual and functional
Minor		
201	Incorrect dimensions, not affecting interchangeability (see 3.3).	SIE
202	Improper workmanship, affecting appearance (see 3.6).	Visual

1/ SIE = Standard inspection equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

4.5.4 QCI failure.

4.5.4.1 Examination. Any relay that fails to conform to any specified requirement shall be rejected, any failure (one or more) of the selected sample in either the Major/Minor categories for the appropriate inspection lot size shall constitute a failure of the entire lot. The rejected relay(s) may be repaired or corrected and resubmitted for inspection. If the contractor utilizes sampling inspection as an element of his inspection, lots may be resubmitted for acceptance if the contractor performs 100 percent inspection on the lot for those characteristics which were defective and resulted in rejection of the lot and removes all defective units or obtains procuring activity approval to resample the lot due to the insignificance of the defects. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots.

4.5.4.2 Tests. Failure of the relay to pass the QCI tests shall be cause for the Government to refuse acceptance of the relay. Any relay containing one or more defects either shall not be submitted for Government acceptance or shall be corrected and retested and resubmitted without defects for Government acceptance. Resubmitted relays shall be kept separate from new relays, and shall be clearly identified as resubmitted relays.

4.6 Control tests. Relays for control tests shall be selected in accordance with 4.6.1, and subjected to the tests specified in table I.

4.6.1 Frequency.

4.6.1.1 Frequency 1 to 50. Control test shall be conducted on one relay from each lot of 50 units consecutively produced, except that not more than one test shall be performed in a one month period, nor less than one test in a three month period.

4.6.1.2 Frequency 1 of 200. Control tests shall be conducted on one relay from each lot of 200 units consecutively produced, except that not more than one test shall be performed in a two month period, nor less than one test in a six month period.

4.6.2 Failure. Failure of any relay to pass any of the specified control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.7 Methods of inspection.

4.7.1 Materials and construction. Conformance to 3.2 and 3.3 shall be determined by inspection of contractor records providing proof or certification that construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industrial standards, test reports and rating data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

4.7.2 Defects. Conformance to 3.3 thru 3.3.5, 3.5, and 3.6, shall be determined by examination for the defects listed in table III. Examination shall be visual, functional, or by measurement with standard inspection equipment.

4.7.3 Insulation resistance. To determine conformance to 3.4.1, measure the insulation resistance between the points specified in 4.7.4 in accordance with MIL-STD-202, method 302, test Condition B. Verify that the resistance is not less than 1000 megaohms.

4.7.4 Dielectric strength. To determine conformance to 3.4.2, perform a dielectric strength test between the specified points of application (except across the normally open contacts) in accordance with MIL-STD-202, method 301. Use 500 vdc/300 vrms across open contacts and 1500 vdc/1000 vrms on all other contacts, for a test period of 60 $\pm$ 2 seconds for initial production inspection. During this test, verify that the leakage current does not exceed 1.0 mA and that the assembly shows no puncture of insulation, no continuing sparkover, and no breakdown of dielectric materials. Points of application:

a. Between case and all contact terminals (2, 3, and 4)

b. Between case and coil terminals (1, 5)

c. Between all contacts (2, 3, and 4) and coil (1, 5)

d. Between normally open contact terminals (2, 4)

4.7.4.1 Acceptance test. The test shall be the same as 4.7.4 except the test period shall be 5 seconds.

4.7.5 Contact characteristics.

4.7.5.1 Rated load. To determine conformance to 3.4.3.1, apply 2.0 A at 28.0 vdc to the assembly contacts for a period of 60 $\pm$ 2 seconds. At the end of this time, the assembly shall show no evidence of physical damage.

4.7.5.2 Resistance. To determine conformance to 3.4.3.2, with no coil power applied, measure the resistance through all closed contacts in accordance with MIL-STD-202, method 307. Each resistance shall not exceed 100 milliohms. Repeat the test with coil power applied.

4.7.5.3 Life. To determine conformance to 3.4.3.3, operate the assembly for 100,000 cycles while energizing the coil with 4 mA for 1 second and de-energizing it for 1 second, and with the contacts carrying 2.0 $\pm$ 0.1 A at 28.0 vdc resistive load. At the conclusion of this test, subject the assembly to the resistance test of 4.7.5.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

4.7.6 Coil characteristics.

4.7.6.1 Current. To determine conformance to 3.4.4.1, apply 18 vdc to the assembly at the coil for a period of 2 ± 1 second. Gradually lower the input voltage to zero and then gradually increase it until the assembly operates. Upon operation (pull-in), the current shall be between 2.5 and 3.6 mA dc. Lower the input voltage until the contacts return to the de-energized position at which time the drop-out current shall be between .36 and 1.8 mA dc. The difference between the pull-in and drop-out current shall be not less than .70 mA dc.

4.7.6.2 Voltage. To determine conformance to 3.4.4.2, apply 18.0 vdc to the assembly at the coil and maintain it for one-half hour without interruption. At the conclusion of this time, subject the assembly to the current test of 4.7.6.1.

4.7.7 DC resistance. To determine conformance to 3.4.5, subject the assembly to the dc resistance test specified in MIL-STD-202, Method 303. The dc resistance of the coil shall be 2000 ± 200 ohms.

4.7.8 Inductance. To determine conformance to 3.4.6, measure the coil inductance using an impedance bridge or equivalent. Inductance shall not exceed 25 H when tested at 60 Hz.

4.7.9 Environmental conditions.

4.7.9.1 Temperature.

4.7.9.1.1 High-temperature. To determine conformance to 3.4.7.1, subject the assembly to the high-temperature test specified in MIL-STD-810, method 501, procedure II, except storage temperature of step 4 shall be 160°F. At the conclusion of the storage temperature test, stabilize the assembly at 145°F and subject it to the performance tests of 4.7.3 thru 4.7.8. After testing, return the assembly to room ambient temperature and subject it to the performance tests of 4.7.3 thru 4.7.8.

4.7.9.1.2 Low-temperature. To determine conformance to 3.4.7.1, subject the assembly to the low temperature test specified in MIL-STD-810, method 502, procedure I. Maintain the temperature at -65°F for a period of not less than 12 hours. At the conclusion of this time, stabilize the assembly at -25°F and subject it to the performance tests of 4.7.3 thru 4.7.8. After testing, return the assembly to room ambient temperature and subject it to the performance tests of 4.7.3 thru 4.7.8.

4.7.9.2 Vibration. To determine conformance to 3.4.7.2, mount the assembly in any axis shown in figure 2 and subject it to the vibration test specified in MIL-STD-810, method 514, procedure VIII, except as follows:

a. Test level shall be the vibration curve shown in figure 1 conducted in a 145°F ambient environment. Time schedule shall be a total of 80 minutes per axis (see figure 2), including 13-1/3 minutes dwell time at each resonance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

b. Connections and instrumentation shall be attached to the assembly to permit operation and testing as specified in c through f.

c. Conduct resonance search at reduced vibration amplitude, at room temperature, and with input power applied.

d. Place the assembly in a temperature chamber and conduct the resonance dwell and sinusoidal cycling tests as specified, with input power applied 50 percent of the time.

e. During vibration, when input power is applied, monitor for intermittent conditions not to exceed 10 ms.

f. After vibration, return the assembly to room ambient temperature and subject the assembly to the performance tests of 4.7.3 thru 4.7.8.

4.7.9.2.1 Control test. This test shall be the same as 4.7.9.2, except the temperature shall be 70 ± 10°F and the time schedule shall be a total of 80 minutes per axis, including 2 minutes dwell time at each resonance.

4.7.9.3 Shock.

4.7.9.3.1 Basic design. To determine conformance to 3.4.7.3, mount the assembly in any axis shown in figure 2 and subject it to the shock test specified in MIL-STD-810, method 516, procedure I, for a total of eighteen half-sine wave shock pulses (six in each axis). Peak amplitude shall be 40 ± 4g with a time duration of 18 ± 3 ms measured at the 10 percent amplitude points. During three of the drops in each axis, apply the input power and monitor for intermittent conditions not to exceed 10 ms. At the conclusion of this test, subject the assembly to the performance tests of 4.7.3 thru 4.7.8.

4.7.9.3.1.1 Control test. This test shall be the same as 4.7.9.3.1 except the shock level shall be limited to 30 ± 3g, 135 ± 13.5g and 285 ± 28.5g.

4.7.9.3.2 High-intensity test. To determine conformance to 3.4.7.3, with the assembly mounted as specified in 4.7.9.3.1, subject it to the shock test specified in MIL-STD-810, method 516, procedure IV for a total of 18 half-sine wave shock pulses (six in each axis). Peak amplitude shall be 180 ± 18g with a time duration of 1.7 ± 0.2 ms in the longitudinal and transverse axes and 380 ± 38g with a time duration of 1.0 ± 0.1 ms in the vertical axis measured at the 10 percent amplitude points. During three of the drops in each axis, apply the input power and monitor for intermittent conditions not to exceed 10 ms. At the conclusion of this test, subject the assembly to the performance tests of 5.7.3 thru 4.7.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

4.7.9.4 Humidity. To determine conformance to 3.4.7.4, place the assembly in a humidity chamber and subject it to the humidity test specified in MIL-STD-810, method 507, procedure II. During exposure, subject the assembly to the performance test of 4.7.5.1. After exposure, subject the assembly to the performance tests of 4.7.3 thru 4.7.8.

4.8 Preparation for delivery. The assembly shall be inspected by the supplier prior to shipment to ensure conformance to the preservation, packaging, packing, and marking requirements of section 5. Sampling inspections shall be limited to those characteristics that can be determined by visual examination.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The armature relay in intended to provide a switching function when used in the Electronic Controller Unit 11608300, a part of the stabilization system of the M60A1 tank.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
c. Whether first article is required (see 3.1).
d. Whether preproduction inspection is required (see 4.4.1).
e. Whether initial production inspection is required (see 4.4.2).
f. Whether test conditions shall be other than as specified (see 4.3).
g. If sampling plan for QCI is other than as specified (see 4.5.1).
h. Selection of applicable level and packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the relay(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4.1 and 4.4.2. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

provide that the Government reserves the right to waive the requirements for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending.

6.4 Definitions.

6.4.1 QCI. The following definitions may be related to the sampling plans specified in section 4:

a. Classification of defects. A classification of defects is the enumeration of possible defects of the unit of product classified according to their seriousness. A defect is any nonconformance of the unit of product with specified requirements. Defects will normally be grouped into one or more of the following classes: critical, major and minor defects. Also, defects may be grouped into other classes, or into subclasses within these classes.

b. Critical defects. A critical defect is a defect that judgement and experience indicate would result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product, or a defect that judgement and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

c. Critical defective. A critical defective is a unit of product which contains one or more critical defects and may also contain major and/or minor defects.

d. Defective. A defective is a unit of product which contains one or more defects.

e. Formation of lots or batches. The product should be assembled into identifiable lots, sublots, batches, or in such other manner as may be prescribed (see 1). Each lot or batch should, as far as is practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time.

f. Lot or batch. The term lot or batch will mean “inspection lot” or “inspection batch”, i.e., a collection of units or products from which a sample is to be drawn and inspected and may differ from a collection of units designated as a lot or batch for other purposes (e.g., production, shipment, etc.).

g. Lot or batch size. The lot or batch size is the number of units of product in a lot or batch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

h. Major defect. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose.

i. Major defective. A major defective is a unit of product which contains one or more major defects, and may also contain minor defects but contains no critical defect.

j. Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.

k. Minor defective. A minor defective is a unit of product which contains one or more minor defects but contains no critical or major defect.

l. Presentation of lots or batches. The formation of the lots or batches, lot or batch size, and the manner in which each lot or batch is to be presented and identified by the supplier will be designated or approved by the responsible authority. As necessary, the supplier will provide adequate and suitable storage space for each lot or batch, equipment needed for proper identification and presentation, and personnel for all handling of product required for drawing of samples.

m. Representative sampling. When appropriate, the number of units in the sample should be selected in proportion to the size of the sublots or subbatches, or parts of the lot or batch, identified by some rational criterion. When representative sampling is used, the units from each part of the lot or batch should be selected at random.

n. Sample. A sample consists of one or more units of product drawn from a lot or batch, the units of the sample being selected at random without regard to their quantity. The number of units or product in the sample is the sample size.

o. Sampling plan. A sampling plan indicates the number of units of product from each lot or batch which are to be inspected (sample size or series of sample sizes) and the criteria for determining the acceptability of the lot or batch (acceptance and rejection numbers).

p. Time of sampling. Samples may be drawn after all the units comprising the lot or batch have been assembled, or samples may be drawn during assembly of the lot or batch.

6.4.2 Recovered materials. “Recovered materials” means materials that have been collected or recovered from solid waste (see 6.4.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

6.4.3 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows, or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.5 Supersession data. This specification supersedes MIL-R-62197(AT), dated 28 October 1983.

6.6 Subject term (key word) listing.

Electromagnetic
Hermetically sealed
Single pole, double throw (SPDT)
Stabilization system
Switch

6.7 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

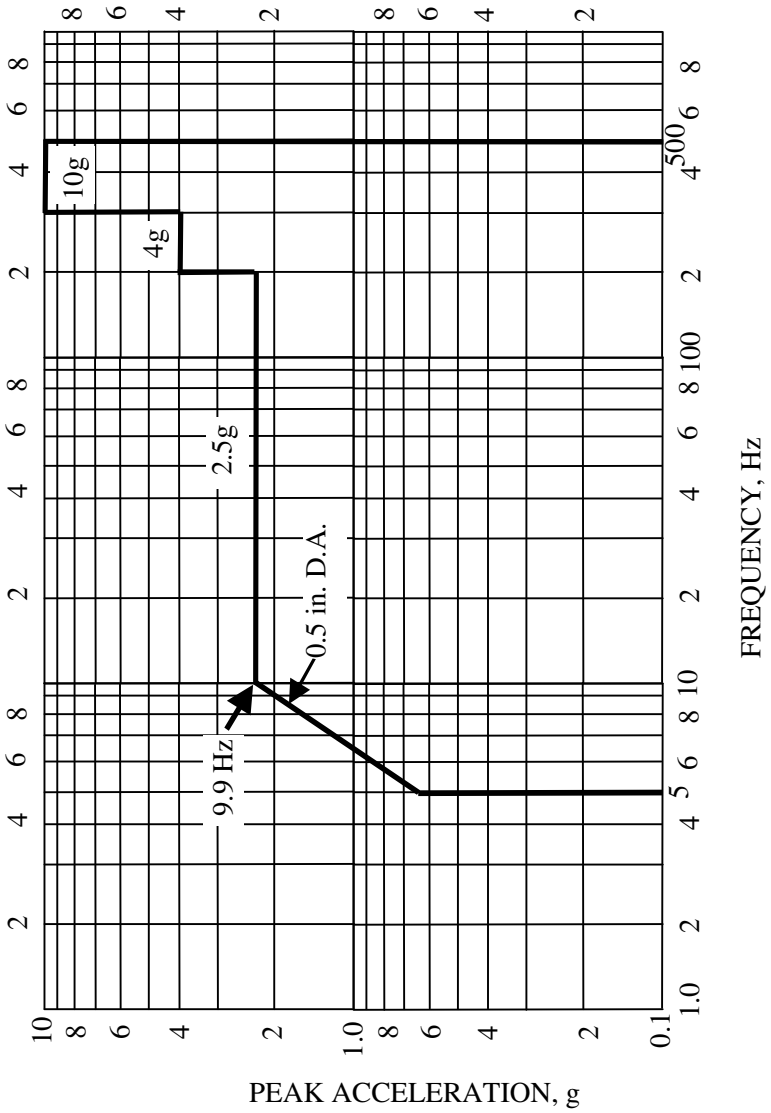


FIGURE 1. Vibration test level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

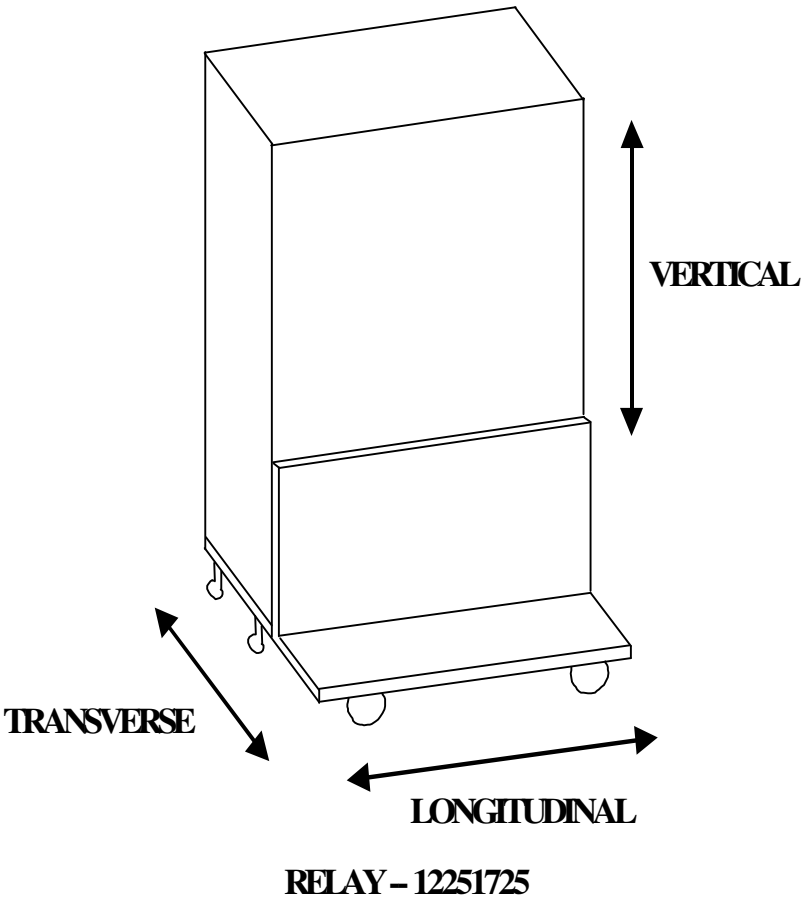


FIGURE 2. Shock and vibration axis definition.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 1250-A022)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:25Z
Check the source to verify that this is the current version before use.